



Assistive Silent Speech Recognition Using Wearable fEMG & Deep Learning

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Project Description

This project proposes a wearable, end-to-end automatic speech recognition system that uses facial electromyography (EMG) signals to decode silent speech intentions into text and audio. The system is designed as a cost-effective assistive technology for individuals with vocal impairments and for communication in silent environments.

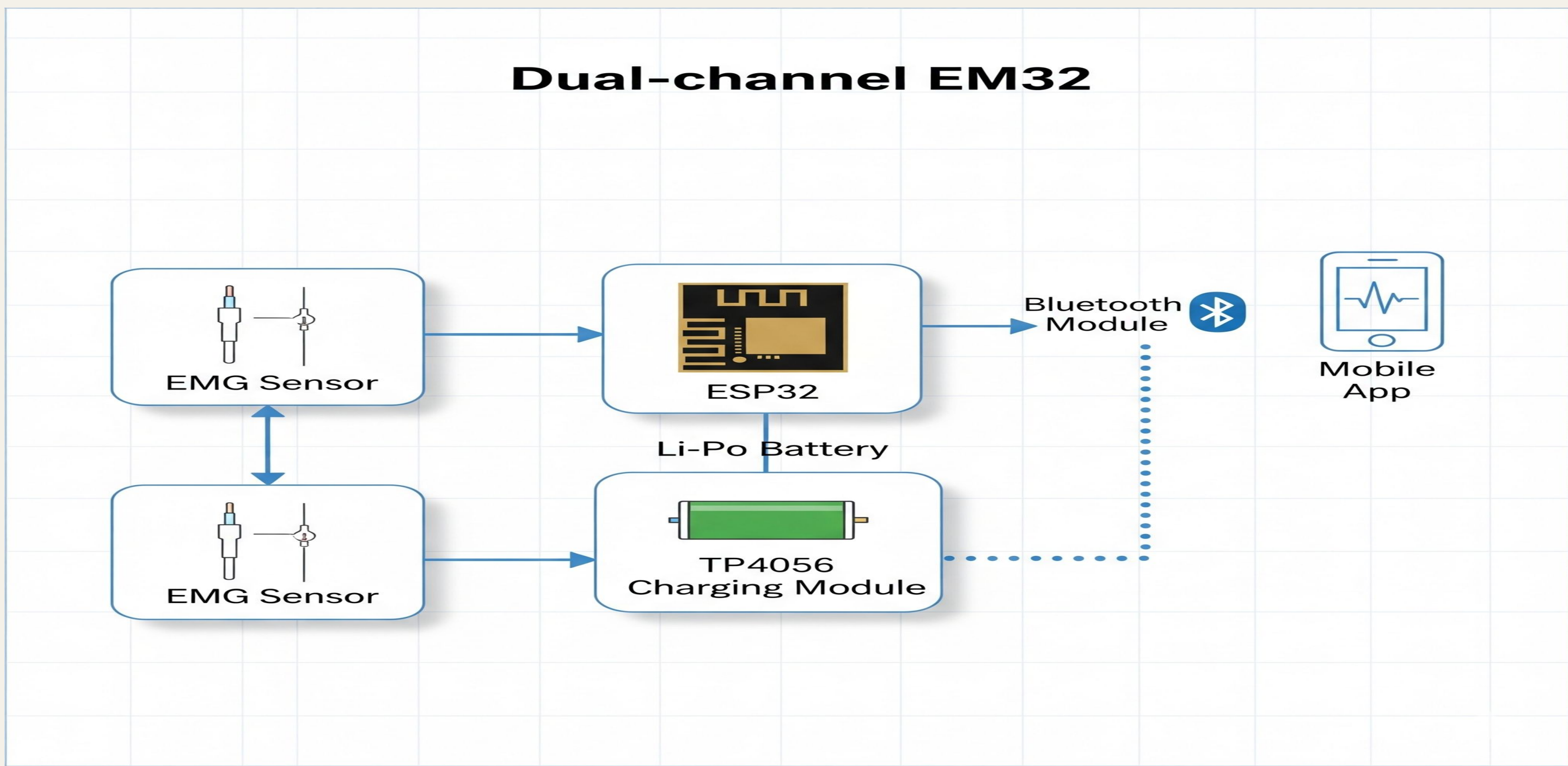
Objectives

- To train a hybrid AI model (GRU + LSTM) for EMG-based utterance classification.
- To integrate multi-modal data (audio, video, fluorescent biomarkers) for improved accuracy.
- To design an end-to-end system with a PCB-level prototype, Bluetooth connectivity, rechargeable battery, and real-time app integration.

Novelty

- We boost recognition by checking fEMG signals against subtle facial cues and timing from audio/video.
- An adaptive end-to-end model makes the speech conversion smooth and reliable.
- A simple Bluetooth app shows live text, speaks it out, and even lets you share on WhatsApp.
- Built-in translation turns your text or speech into different languages instantly.

Circuit diagram/flowchart/algorithm/design



Results/Table/Design equations

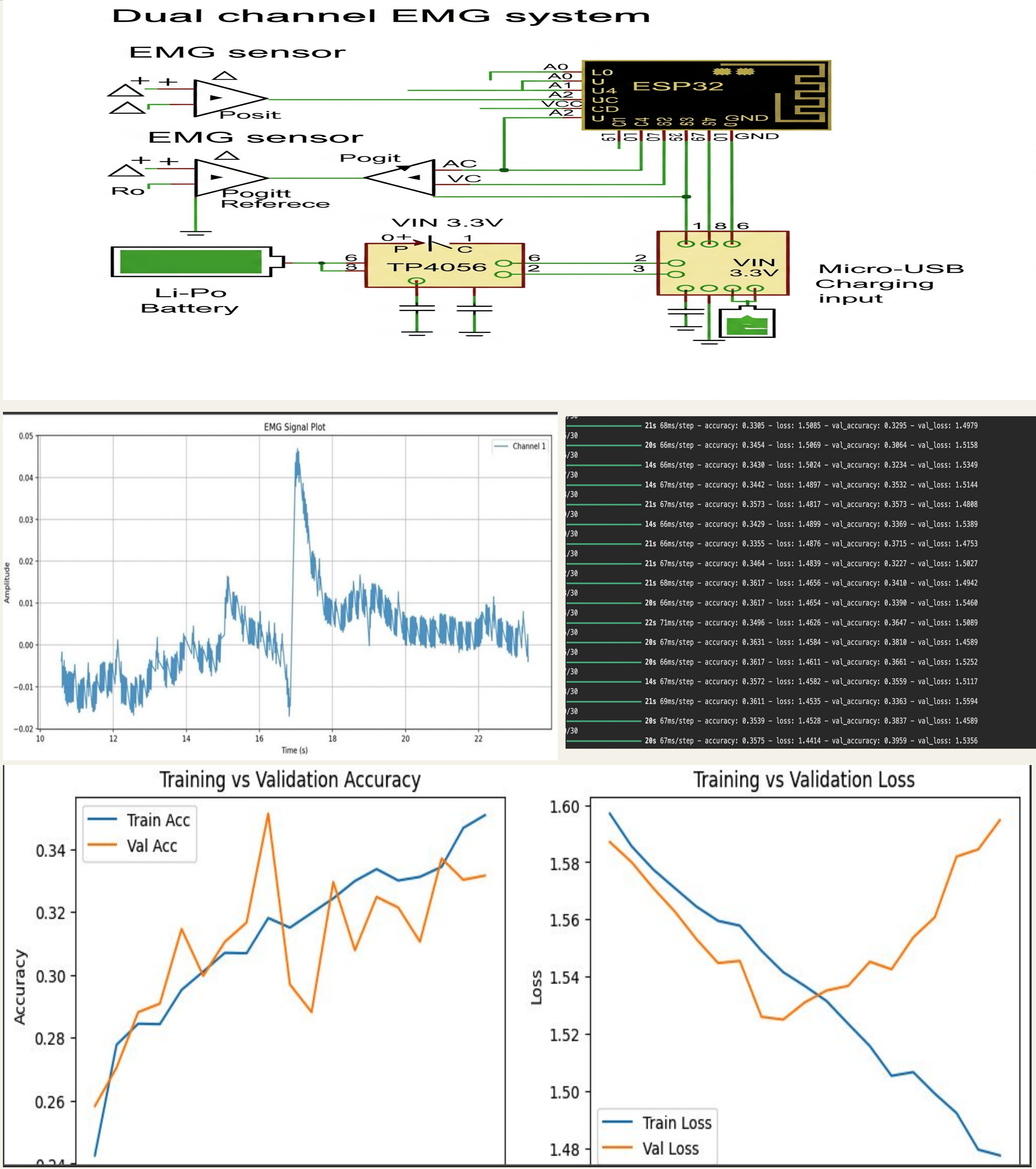
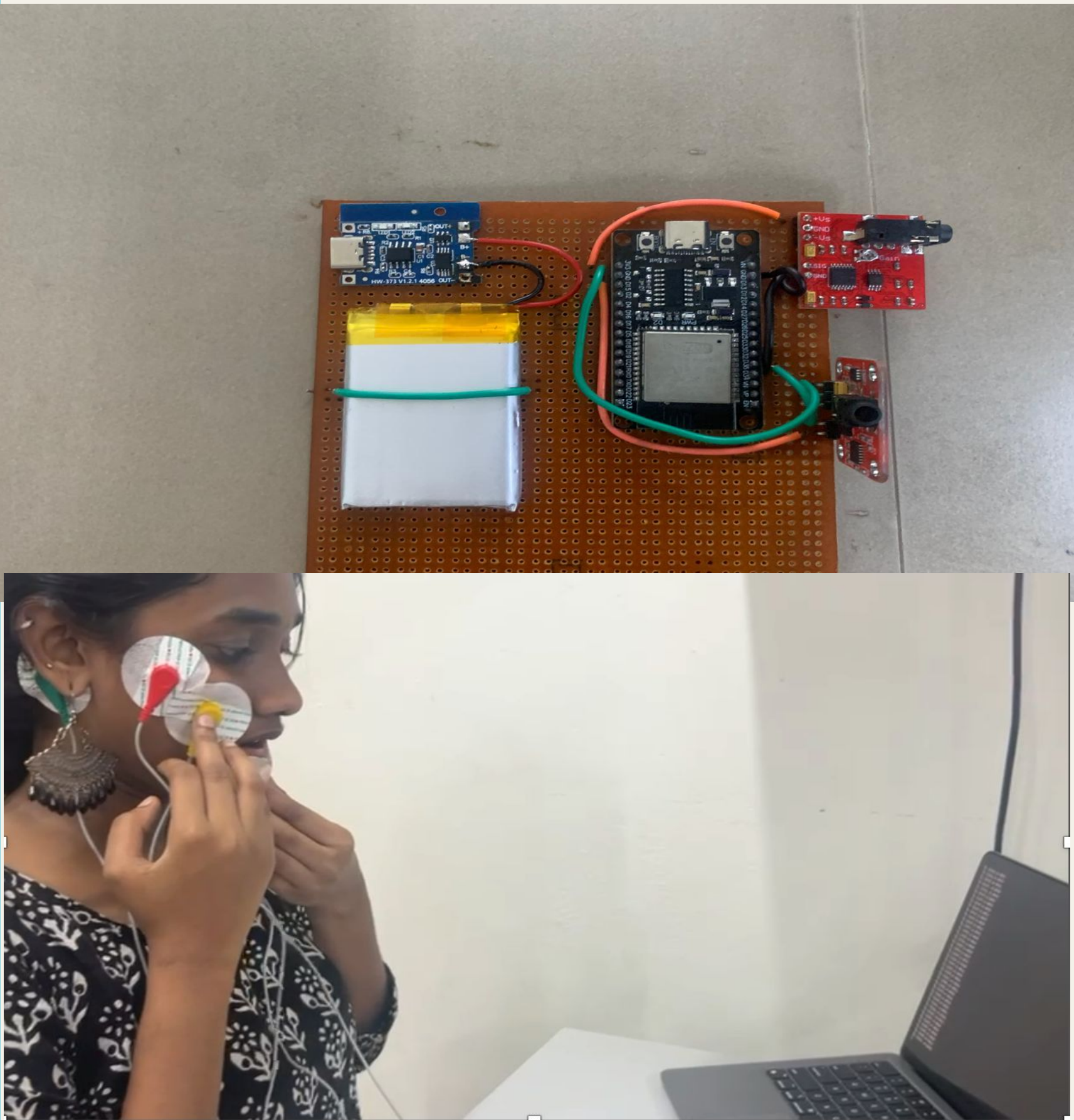


Photo of the Project



Outcome of the Project

Impact on Industry/Society

- Wearable assistive device for speech-impaired.
- Enables silent communication in noisy settings.
- Useful in rehab for stroke, ALS, vocal cord injury.
- Low-cost, portable, and practical solution.
- Promotes inclusivity and better quality of life.

Conclusion

- Successfully demonstrated a wearable EMG-based silent speech recognition system.
- Achieved high accuracy in mapping EMG signals to basic speech commands.
- The prototype is low-cost, portable, and practical for real-world use.
- Can be further extended with AI/Deep Learning for continuous speech recognition.

References

1. https://www.isca-archive.org/interspeech_2018/diener_18_interspeech.pdf
2. <https://arxiv.org/abs/2504.13921>
3. Meltzner GS, Heaton JT, Deng Y, De Luca G, Roy SH, Kline JC. Development of sEMG sensors and algorithms for silent speech recognition. J Neural Eng. 2018 Aug;15(4):046031. doi: 10.1088/1741-2552/aac965. Epub 2018 Jun 1. PMID: 29855428; PMCID: PMC6168082.